

□CaptionChameleon

Adaptive Multi-Style Video Captioning at Scale

AMD Hackathon 2024

The Challenge

- Video content grows exponentially every year
- Different audiences require different caption styles
- Manual captioning is expensive and time-consuming
- Traditional ML approaches require extensive labeled data
- Scaling multi-style caption generation is complex

Our Solution

Zero-Shot Learning + Style-Conditioned Prompts

- Generate all 4 caption styles in ONE API call per video
- No expensive model fine-tuning required
- Parallel processing delivers 10x throughput
- Production-ready Docker containerization
- Cost-effective scaling to thousands of videos

Four Perfect Styles

Formal

Professional, objective descriptions for technical documentation and formal contexts.

Sarcastic

Dry, witty commentary with ironic undertones for entertainment and social media.

Tech Humor

Programming and DevOps humor for technical audiences and developer communities.

Casual

Everyday humor accessible to general audiences without technical jargon.

How It Works

- **Frame Extraction:** Sample 8 frames intelligently from each video
- **Batch Processing:** Generate all 4 styles in a single API request
- **Parallel Execution:** Process 10 videos simultaneously
- **Vision-Language Model:** Qwen3-VL 8B with zero-shot capabilities
- **Output:** JSON with formatted captions for each style

Performance Metrics

~15s

3 videos processed

570MB

Image size (< 10GB limit)

4x

Styles per API call

10

Parallel workers

Production Ready

- **Public Docker Image:** shadowking9021/amd-hackathon:latest
- **Platform Verified:** linux/amd64 manifest certified
- **Size Optimized:** 570MB (well within 10GB limit)
- **Fully Documented:** Comprehensive README and examples
- **Scalable:** Easily adjust parallelism and frame count

□ Ready to Deploy

Docker Image:

`shadowking9021/amd-hackathon:latest`

Process hundreds of videos • Get 4 perfect captions • Zero fine-tuning required

Thank You

CaptionChameleon: Adaptive Multi-Style Video Captioning

Questions? Check out the GitHub repository for complete documentation and examples.

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